January 2026

IPC Online

Section: A2

Time: 30 min

Run two threads.

a. Thread1 prints ‘0 ’ in an infinite loop one at a time
b. Thread2 prints ‘1 ’ in an infinite loop one at a time

Now, these two threads should run in a that at any point in time, the difference between the number of ‘0’s printed and the number of ‘1’s printed is at most 2.

The output may look like 0 1 1 1 0 0 0 0 1 …

Implement this using two counting semaphores only.

Create a file <student_id>.cpp and submit it.